Wideband High Directive Aperture Coupled Microstrip Antenna Design by Using a FSS Superstrate Layer

Abbas Pirhadi, Hadi Bahrami, and Javad Nasri

***Abstract*—in this communication a high directive electromagnetic bandgap (EBG) antenna operating in wide frequency band for both return loss (RL) and directivity is examined. In this EBG antenna an aperture coupled microstrip antenna (ACMA) is used as a feeding source and a frequency selective surface (FSS) is used as a superstrate layer. Suitable use of the superstrate layer, microstrip patch and coupling aperture simultaneously, leads to produce separate resonance frequencies and therefore the wide frequency band for RL. Also, high directivity is achieved only by using the superstrate layer that has been made by the FSS layer with square loop elements. At first, a wideband ACMA is designed to operate in x-band. In this step appropriate design of coupling aperture is of a great importance. Secondly, after the design of optimum superstrate layer by the FSS structure, it is added to the ACMA in order to increase both bandwidth and directivity.**

***Index Terms*—Electromagnetic bandgap, frequency selective surface (FSS), microstrip antennas, superstrate layer.**

I. Introduction

Electromagnetic bandgap (EBG) structures are periodic structures that can be used to control and manipulate the propagation of electromagnetic waves. The high directive resonator antenna is an example of antenna application of EBG structures which is also called EBG antenna [1]–[3]. The EBG antenna consists of a feeding source and some superstrate layers. By suitable design of the superstrate layers it is possible to obtain a high directive antenna. There are several configurations for the design of superstrate layer. In low profile antenna applications this layer must be compact, easy to fabricate and commercially available [4]–[7]. To this purpose, frequency selective surfaces (FSS) can be good candidates. Various configurations such as ring, patch, square loop, strip and slot shapes have been used to design of the FSS superstrate layer [8]–[10].

By using the FSS superstrate layer singularly, or its combination with a reactive surface, the bandwidth, polarization and radiation pattern characterizations such as side lobe level (SLL) can be controlled [11]–[15]. Beside the enhancement of directivity, the FSS superstrate layer can be used as a polarizer. In this case, it is possible to obtain circular polarization by using a single source [16], [17].

One of the most important challenges in the design of the EBG antenna is its frequency bandwidth for the return loss and maximum directivity. These characteristics depend on the FSS superstrate layer and feed source. Most of the works that have been done emphasize on achieving wide-frequency band or multi-frequency band by a suitable design of FSS superstrate layer [6], [11], [14], [15], [18], [19]. A few works have been done on the feeding source of the EBG antennas, especially for the bandwidth of the return loss of the EBG antenna. Among

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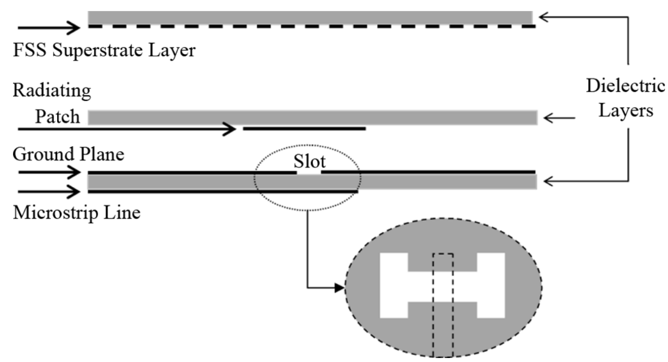

Fig. 1. Geometry of EBG antenna fed by aperture coupled microstrip patch.

various feeding sources that have been used, microstrip antennas, because of their low profile and easy construction are very desirable. The authors in [20], beside the enhancement of directivity, have studied the effect for the FSS superstrate layer on the return loss of a probe-fed microstrip antenna and have exhibited how the FSS superstrate layer can improve the bandwidth of such antenna.

In this communication by using an aperture coupled microstrip antenna (ACMA) as the feeding source and an FSS layer as the superstrate, the EBG antenna will be studied. The return loss of ACMA has the widest frequency bandwidth among the single fed microstrip antennas. This phenomenon is obtained by the suitable design of coupling aperture configuration, but the directivity of antenna decreases in this frequency band. The FSS superstrate layer is used to solve this problem. In addition, beside the directivity, a wide frequency band for the return loss by a suitable design of the FSS superstrate layer is obtained.

In Section II, the FSS superstrate layer and ACMA are designed separately and in Section III, the combination of ACMA and FSS superstrate layer are studied. In this section the most effective parts that influence the radiation characteristics and the bandwidth of antenna, are examined parametrically. In Section IV, the leaky modes in the EBG antenna which are excited due to the surface current of the patch, are studied. Also, in this section a suitable circuit model for the antenna is proposed. Finally, the simulations are verified with experimental results. All simulations have been done by ANSOFT HFSS software based on FEM.

II. Description of Antenna Configuration

The antenna structure is composed of three sections including coupling aperture, radiating patch and the FSS superstrate layer (Fig. 1).

A. Design of the FSS Superstrate Layer

There are different configurations and methods to design the superstrate layer. In fact, this layer acts as a partially reflecting surface. Near its resonance frequency where the reflection coefficient of surface is about unit, the radiating source and the superstrate layer produce resonance condition in which the directivity of the antenna increases considerably [6], [10].

The resonance frequency of the FSS superstrate layer has to coincide in operating frequency.

In this communication the square loop elements have been used to design the FSS superstrate layer. The square loop has symmetrical geometry and can be used in dual polarize and dual band applications. Also, compared with the patch element it has a more compact size and a more controllable resonance frequency.

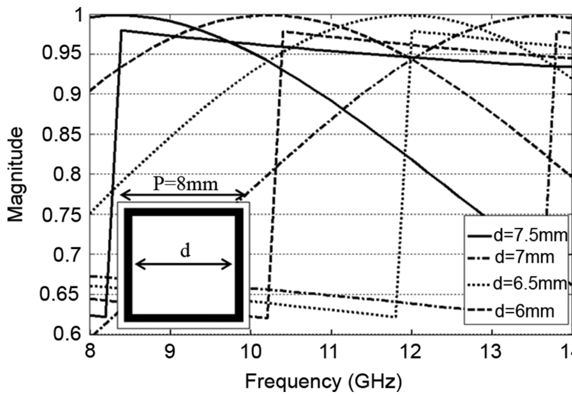

Fig. 2. Magnitude and phase of reflection coefficient of FSS.

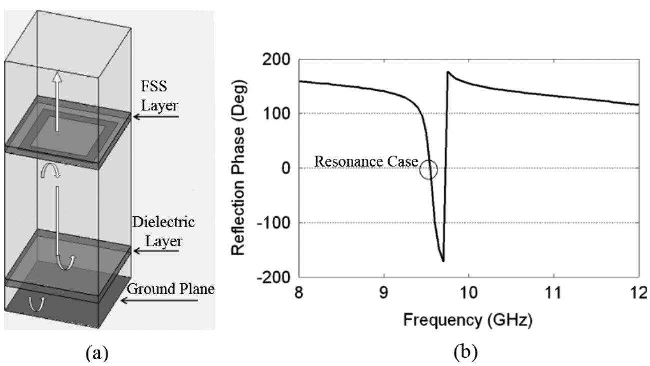

Fig. 3. (a) Unit cell of the whole structure (b) Phase of the reflection coefficient for the resonance condition of unit cell.

Fig. 2 shows the amplitude and phase of reflection coefficient of the FSS superstrate layer . Because of the periodicity of the FSS superstrate layer, to determine its reflection coefficient, it is enough to analyze just one cell.

Fig. 3 shows how the FSS superstrate layer in combination with an aperture coupled microstrip antenna is used to produce resonance condition. In this configuration, there are three layers that are important in producing resonance condition. These layers are the ground plane in which the coupling aperture of ACMA has been placed, the dielectric layer in which the patch has been placed, and the FSS superstrate layer.

The reflection coefficient of the ground plane for horizontal electric fields is (-1). The dielectric layer reflects and transmits the EM fields with coefficients proportional to the thickness and dielectric properties of the layer and finally as mentioned the FSS layer reflects and transmits the EM fields according to the Fig. 2. Among these layers, only the FSS layer can be adjusted. This layer must be adjusted so that all of the EM fields in outer region of the structure are added constructively. In this case, the reflection coefficient phase of the whole structure in Fig. 3 is $2n\pi$ and the resonance condition takes place [24].

The most important parameters to adjust the resonance frequency are the height of the FSS layer, the periodicity of the FSS elements, and their dimensions. In the next section the effects of these parameters on the antenna characteristics are studied.

B. Design of Wideband Aperture Coupled Microstrip Antenna

The configuration of ACMA is shown in Fig. 4. Many useful features of this type of antenna attract designers to use it for various applications [21]–[23].

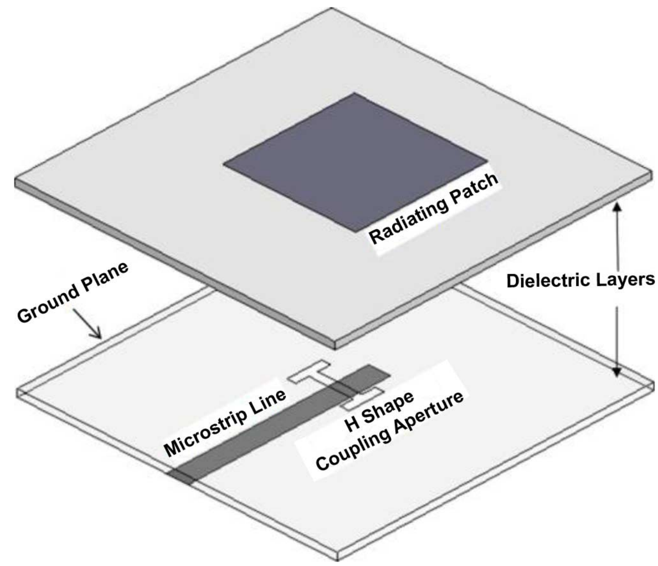

Fig. 4. Configuration of ACMA.

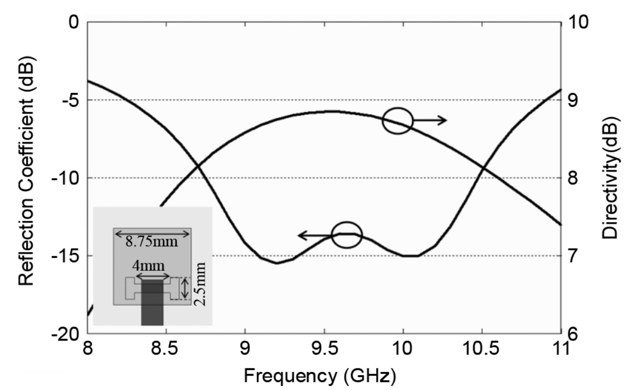

Fig. 5. Return loss and directivity of ACMA fed by H shape slot and $\text{dielectric height} = 3\ \text{mm}$.

One of the useful features of this antenna is its capability to obtain wide bandwidth by suitable design of its coupling aperture. This phenomenon is obtained by using a cross-shaped coupling slot, two orthogonal offset slots and so on. In this communication the H shape slot is used [24]. By using this structure, two resonance frequencies due to the patch and slot are adjusted so that a wide frequency band for return loss is obtained. Fig. 5 shows the directivity and return loss of an ACMA feeding by H shape slot, operating in x-band. The antenna is designed on PCB microwave board (Rogers RO3003) with $\varepsilon_r = 3.38$ and the $\text{thickness} = .782\ \text{mm}$.

By using the H shape slot the resonance frequency of coupling aperture moves next to the resonance frequency of the patch.

III. Design of the Wideband High Directive ACMA and Experimental Results

A. Circuit Model of EBG Antenna

Combination of the ACMA and FSS superstrate layer leads to enhancement of both directivity and bandwidth of antenna. In fact the FSS superstrate layer acts as a parasitic load in multilayer stack patch antennas with low profile configuration. Based on the references [20],

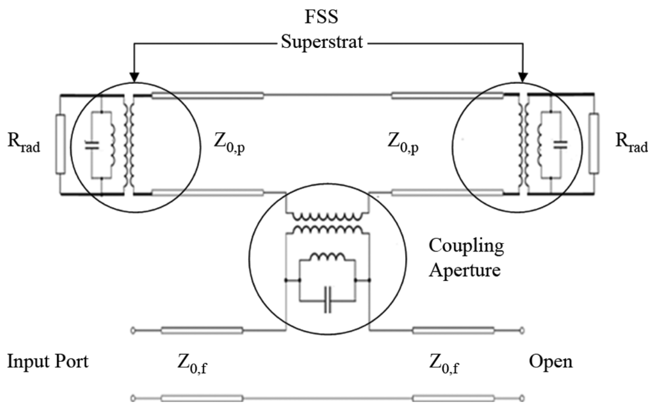

Fig. 6. Circuit model for the ACMA with the FSS superstrate layer.

[25], circuit model for the ACMA with superstrate layer can be demonstrated as follows.

As it is seen in the circuit model, the coupling aperture and the FSS superstrate layer are modeled by resonating circuits. The patch antenna has been depicted by three ports network which has been terminated by these resonating circuits. Therefore, it is expected to have three resonance frequencies due to coupling aperture, the FSS layer, and the Patch. Length of the patch, dimensions of the coupling aperture, height of the FSS layer, dimensions and period of the FSS elements are the important parameters to control the different resonance frequencies. Hence, this configuration has a good potential to achieve the wideband operating frequency, especially for return loss.

B. Parametric Study of Antenna

In this section the effects of important parameters such as the height and dimensions of the FSS layer on both of the RL and directivity of the antenna are studied. By varying the height of the FSS the resonance frequency of the EBG antenna is displaced. Moreover, to keep the resonating condition in the EBG antenna, it is necessary to modify the dimensions of FSS elements. This modification is considerable for large variations of the height of FSS but it is negligible for small variations.

In the following the effects of the height of FSS superstrate layer on the RL and directivity of antenna are shown. To this purpose the EBG antenna is simulated for the different values of FSS superstrate layer heights.Fig. 7(a) depicts that increasing the height of the FSS, leads to the movement of the 2nd resonance frequency toward the lower frequency and also, from Fig. 7(b) the maximum directivity of the antenna move to lower frequency.

Another important parameter in the design of EBG antenna is the FSS dimensions. In this case the antenna is considered as a leaky wave antenna (LWA) in which the leaky waves propagate toward the edges of FSS and produce radiating current distribution on the FSS layer. Increasing the dimensions of FSS layer, leads to the increase of the radiating aperture of the antenna.

In the 2-D LWAs which are constructed using the FSS surface, the leakage current is tapered toward the edges of the FSS [25], [26]. Therefore, the dimensions of the FSS superstrate layer must be chosen such that, the reflected wave from the edges of FSS can be neglected. To examine the effects of the FSS dimensions, the FSS superstrate layer is studied with different number of cells. Fig. 8 depicts the RL and directivity of the antenna for the FSS with different number of cells . Simulation results show that enhancing the dimension of FSS more than 7×7 elements, has a little effect on the $\mathrm{RL} < -10\ \mathrm{dB}$ and width.

To show how the leaky wave propagates in the EBG antenna and also to detect the different modes in the EBG antenna it is necessary to study

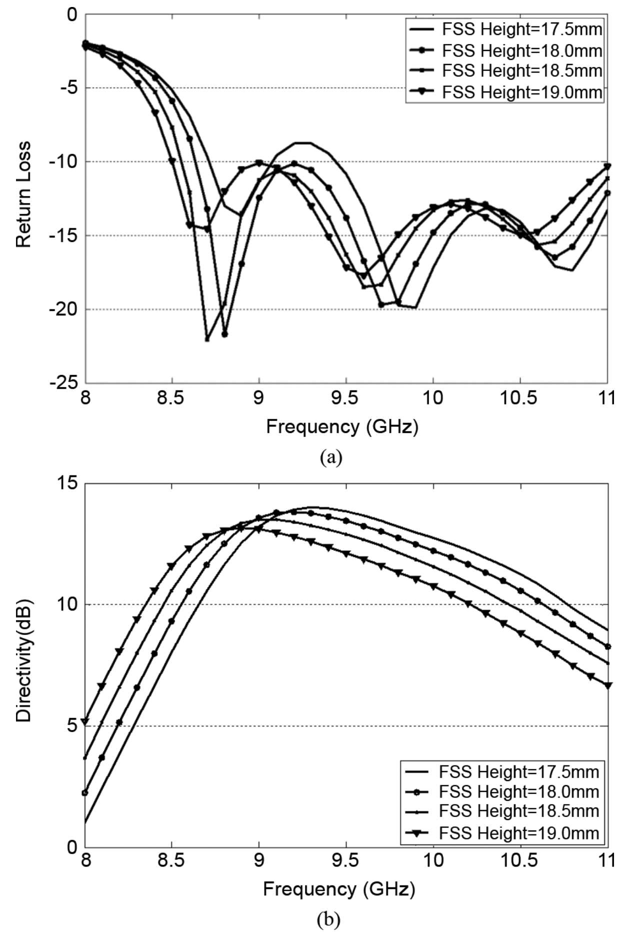

Fig. 7. Effect of FSS height (a) Return loss of EBG antenna (b) Directivity of EBG antenna.

the excitation current in the EBG and also the leakage surface current from the EBG antenna. According to [27], [28], to have a broadside radiation pattern from the 2D LW antenna, it is necessary to excite the suitable horizontal current in the EBG antenna. In this case both TE and TM modes of cavity will be excited. The excitation current in our EBG antenna is the surface current on the patch antenna. This horizontal surface current is directed along the feeding transmission line of ACMA.

Fig. 9 shows the surface current on the patch to excite TE and TM modes of the EBG antenna and also the leakage current on the FSS superstrate layer which produces the radiating Fields . The FSS superstrate layer is 9×9 elements. The periodicity of the FSS elements is 7 mm and the width of square loops of the FSS are 5 mm with 0.2 mm thickness. The simulation of this structure has been done with HFSS software with about 50000 meshes generated by a 2.4 GHz, Quad-Core Processor. As it is seen this current is tapered toward the edges of the cavity. For our design the surface current at the edges of the FSS is about 0.1 of its maximum in the middle of the FSS.

IV. Final Design of Antenna and Measurement Results

Fig. 10 shows the simulation and experimental results of the optimized structure to produce three resonance frequencies and consequently, the wide frequency band.

There are three resonance frequencies in the return loss of the antenna where the first and third resonance frequencies are due to the coupling aperture and patch antenna and the second resonance frequency is due to the FSS superstrate layer. In the Fig. 11 the simulation and

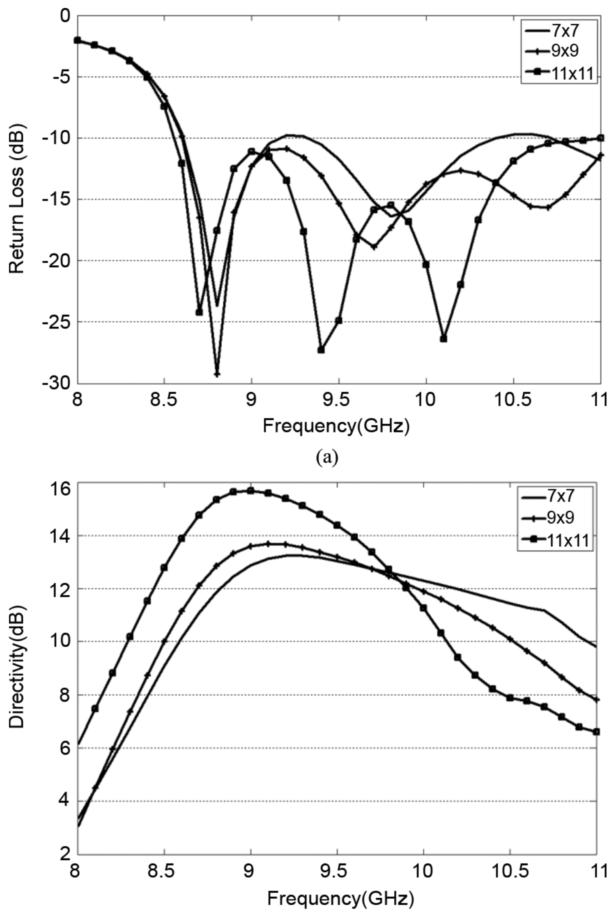

Fig. 8. Effect of the FSS with different cell number (a) return loss of EBG antenna and (b) directivity of EBG antenna.

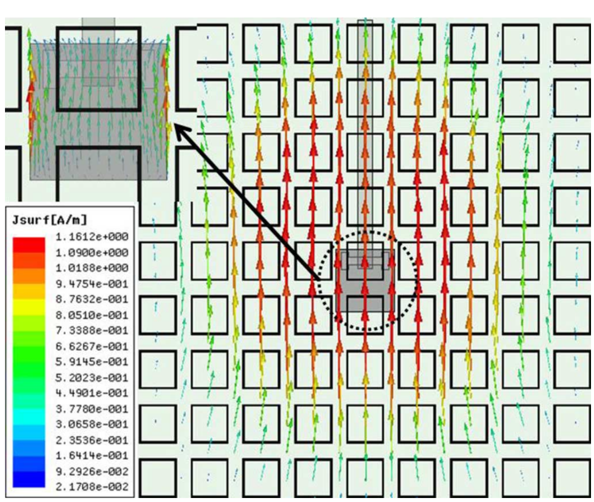

Fig. 9. Current distribution on the patch surface and FSS surface of antenna.

experimental results of the antenna gain has been depicted. The frequency shift between the experiment and simulation results in Figs. 10 and 11 may be due to the fabrication error and experimental effects, but it must be noticed that the expected behavior of the simulation and experimental results are completely achieved in the operating frequency band.

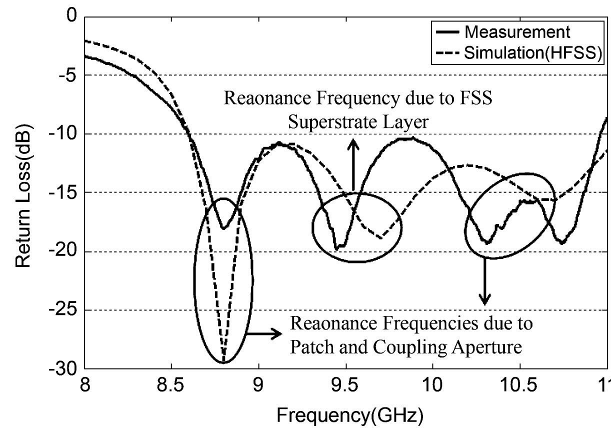

Fig. 10. Return loss of ACMA in the presence of the FSS layer.

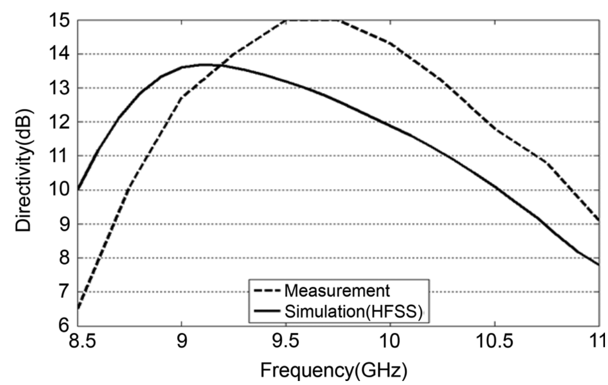

Fig. 11. Directivity of ACMA in the presence of FSS layer.

The bandwidth of the $\mathrm{RL} < -10\ \mathrm{dB}$ of ACMA in Fig. 5 is (8.7–10.5) GHz while this frequency band for the designed EBG antenna in Fig. 10 is (8.6–11) GHz. Therefore, compared with ACMA the bandwidth of the EBG antenna has increased about 5%. Also, at center frequencies the directivity of the EBG antenna is about 15 dBi while the directivity of ACMA is about 9 dBi that shows the directivity of the ACMA antenna has increased about 70% by using the FSS superstrate layer.

The antenna configuration is shown in Fig. 12. In the final design, the air gap between the substrates of the slot and radiating patch is 3.5 mm and the distance of the substrate of the FSS from the patch is 14.75 mm. Also the periodicity of the FSS elements is 7 mm and the width of square loops of the FSS is 5 mm with 0.2 mm thickness.

The Co and Cross polarization radiation pattern of the EBG antenna for different frequencies in E-Plane and H-Plane are shown in the Fig. 13.

The high back lobe level of the antenna is mostly due to the coupling aperture (H shape slot in the ground plane) and can be reduced by putting an absorber material or metallic reflector plate behind the antenna.

V. Conclusion

In this communication, the FSS layer has been used as the superstrate layer for ACMA antenna. Because of the controllable performance of the FSS, it is possible to design the microstrip antenna with more parameters to adjust the antenna characteristics such as return loss and directivity. In the conventional microstrip antennas with dielectric superstrate cover, only the dielectric properties of the superstrate and its

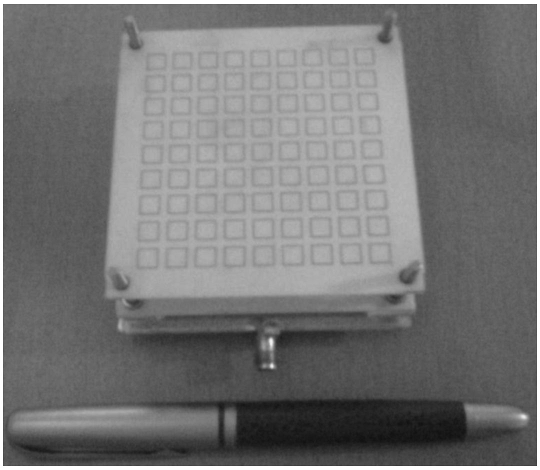

Fig. 12. The ACMA with the FSS superstrate layer.

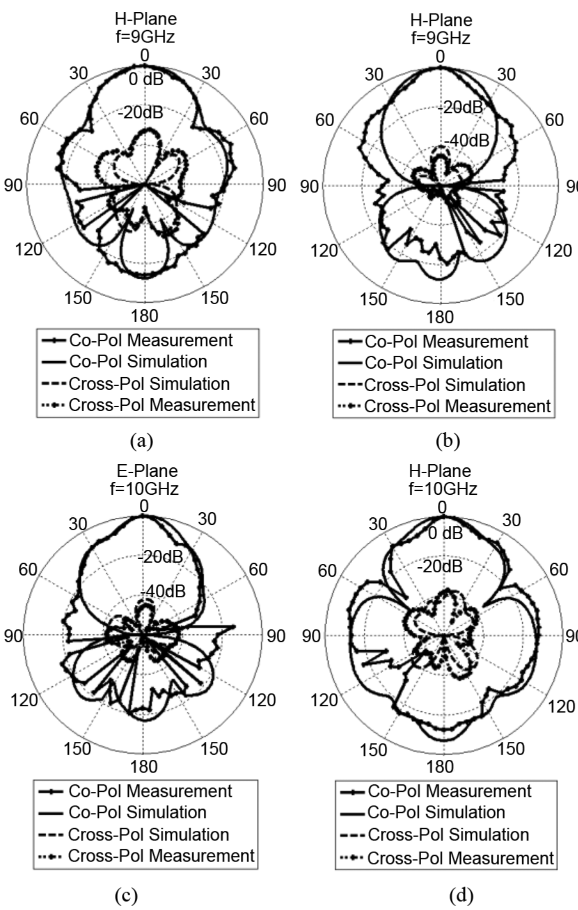

Fig. 13. Measurement and simulation results of Co and cross polarization of electric field radiation patterns for E- and H-planes at 9 GHz and 10 GHz.

thickness are used to achieve the desired characteristics but in the superstrate FSS layer, the FSS has more effective parameters to design of antenna. Also, the combination of the superstrate FSS layer with different types of microstrip antenna such as aperture coupled microstrip antenna helps us to solve some problems of these antennas such as reduction of its directivity in operating frequency band.

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Synthesis of Array Radiation Pattern Footprints Using Radial Stretching, Fourier Analysis, and Hankel Transformation

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***Abstract*—For the synthesis of radiation patterns using antenna arrays with thousands of elements, the combination of direct methods with non-stochastic optimization can be more efficient than stochastic optimization. Here we describe an efficient method for the synthesis of footprint patterns that combines Hankel transformation with Fourier analysis following angle-dependent homothesis (i.e., radial stretching and/or shrinking) of an axisymmetric Elliott-Stern pattern for a circular aperture. The new method can in principle be applied to arrays of radiating elements with arbitrary locations and element patterns.**

***Index Terms*—Antenna arrays, antenna radiation pattern synthesis, footprint patterns.**

I. Introduction

The advent or anticipation of antenna arrays with hundreds, thousands or even millions [1] of elements has changed the priorities of array synthesis methodology. Whereas not long ago designers generally strove to elicit the maximum performance from a limited number of elements by means of sophisticated stochastic optimization techniques [2], satisfactory performance can now often be obtained basically by simply increasing the number of elements, so long as this is not prevented by constraints on antenna size. Very large arrays make exquisite optimization not only unnecessary, but counterproductive, since computation time increases rapidly with array size. In the synthesis of such large arrays, there has accordingly been a trend to return to direct methods, or fast non-stochastic optimization, and to resume exploration of the possibilities of variants of such methods.

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In this context, we have recently been exploring one approach to one of the fundamental aspects of the array synthesis problem: the fact that ideally desirable radiation patterns generally feature discontinuities and/or absolutely flat regions, and can therefore only be approximated by physical systems. Because of this, in choosing among different synthesis methods one is implicitly choosing a particular kind of approximation to the ideal pattern. The approach we have been investigating is based on the intuition that in applying a standard synthesis technique it is in many cases better not to aim directly to approximate the ideally desired pattern, but rather to approximate an intermediate approximation that already possesses some of the characteristics of a physically realizable radiation pattern, such as a radiation-pattern-like lobe structure.

In applying this strategy to the synthesis of irregularly shaped footprints (i.e., flat-topped beams with irregular boundaries), we use the Elliott-Stern (ES) method for circular apertures [3] to obtain an approximately flat-topped circular beam of appropriate size which we then deform by angle-dependent homothesis (i.e., angle-dependent radial stretching or shrinking) to fit the desired footprint boundary. It is this deformed ES beam, $F_{\rm tar}$, that we use as the target of a subsequent direct synthesis method. In previous work, this target has been synthesized by the Woodward-Lawson (WL) method (with iteration of ES-WL cycles) [4], or by least-squares optimization of the excitations of an irregular planar array so as to obtain a pattern fitting a set of samples of $F_{\rm tar}$ [5]. Here we report that very acceptable results can be obtained by approximating $F_{\rm tar}$ by means of a method based on approximate inverse Hankel transformation.

II. Method

Given an ideal footprint defined by a contour C that is star-shaped with respect to the zenith $\theta = 0$ of spherical coordinates (θ, ϕ) (i.e., any point of C can be joined to the zenith by a straight line that does not pass through any other point of C); given a planar array of maximum diameter $2a$ that is to generate the footprint; and given specifications R_d, SLL_d and DRR_d for the ripple R and sidelobe level SLL of the pattern and for the dynamic range ratio of the array ($\mathrm{DRR} = |I_{\max}/I_{\min}|$, where $I_{\max}$ and $I_{\min}$ are respectively the greatest and least excitation amplitudes of the array elements); we proceed as follows.

Step 1. We begin by deforming C to a circle C' around the origin: $(\theta, \phi) \to (\theta', \phi)$, where $\sin\theta' = h(\phi)\sin\theta$ for some positive function h defined on $[0, 2\pi)$. To minimize distortion introduced by subsequent operations, C' should be as close to C as possible.

Step 2. We apply the ES method [3] to obtain a ϕ-symmetric footprint of contour C', maximum tolerable peak-to-trough ripple R_t (for some R_t such that $|R_t| \le |R_d|$) and maximum tolerable sidelobe level SLL_t (for some $\mathrm{SLL}_t \le \mathrm{SLL}_d$) that could be generated by a circular aperture of radius a. That is, we find t_n and s_n such that the required circular footprint $F_{\rm ES}$ is given by a scaled distortion of J_1, the Bessel function of the first kind and order 1:

$$F_{\rm ES}(t,\phi) = 2\frac{J_1(\pi t)}{\pi t} \times \frac{\prod_{n=1}^{M}\left(1-\frac{t^2}{(t_n+js_n)^2}\right)\prod_{n=M+1}^{M+m}\left(1-\frac{t^2}{t_n^2}\right)}{\prod_{n=1}^{M+m}\left(1-\frac{t^2}{\gamma_{1n}^2}\right)} \quad (1)$$